

Digital Signal Processing

Detailed Exam Solutions (Part I - 2023)

Generated Solution Document

Introduction

This document provides complete, step-by-step solutions for the questions in **PART I** of the Digital Signal Processing exam paper (Ref No: Ex/EE/PC/B/T/224/2023).

Question 1

(a) Continuous-Time Signals from Sampled Sequence

Problem: Consider the discrete-time sequence $x[n] = \cos(n\pi/8)$. Find two different continuous-time signals that would produce this sequence when sampled at a frequency of $f_s = 10$ Hz.

Solution:

A continuous-time signal $x(t) = \cos(2\pi f_0 t)$ sampled at intervals $T = 1/f_s$ yields the discrete sequence:

$$x[n] = x(nT) = \cos(2\pi f_0 n / f_s)$$

Equating this to the given sequence $x[n] = \cos(\frac{\pi}{8}n)$:

$$\frac{2\pi f_0}{f_s} = \frac{\pi}{8} \implies f_0 = \frac{f_s}{16} = \frac{10}{16} = 0.625 \text{ Hz}$$

Thus, the first continuous-time signal is:

$$x_1(t) = \cos(2\pi(0.625)t) = \cos(1.25\pi t)$$

Due to aliasing, any continuous-time frequency $f_k = f_0 + kf_s$ (where k is an integer) will produce the exact same discrete-time samples. Let $k = 1$:

$$f_1 = 0.625 + 10 = 10.625 \text{ Hz}$$

Thus, a second continuous-time signal is:

$$x_2(t) = \cos(2\pi(10.625)t) = \cos(21.25\pi t)$$

(b) Nyquist Sampling Rates for Derived Signals

Problem: If a bandlimited continuous-time signal $x(t)$ has a Nyquist sampling rate of ω_s , what is the Nyquist sampling rate for: (i) $x(3t)$ (ii) $x^3(t)$ (iii) $2x(t)\cos(\omega_0 t)$. Justify.

Solution:

The Nyquist sampling rate is exactly twice the maximum angular frequency present in the signal: $\omega_s = 2\omega_{max}$. Thus, the maximum frequency of $x(t)$ is $\omega_{max} = \omega_s/2$.

- (1) $x(3t)$: Time scaling by a factor of 3 compresses the signal in time, which expands it in the frequency domain by a factor of 3. The new maximum frequency is $3\omega_{max} = 1.5\omega_s$.
Nyquist Rate: $2(1.5\omega_s) = 3\omega_s$.
- (1) $x^3(t)$: Multiplication in the time domain corresponds to convolution in the frequency domain. Convolution of a spectrum bandlimited to ω_{max} with itself three times expands the total bandwidth by a factor of 3. The new maximum frequency is $3\omega_{max} = 1.5\omega_s$.
Nyquist Rate: $2(1.5\omega_s) = 3\omega_s$.
- (1) $2x(t)\cos(\omega_0 t)$: Multiplying by a cosine modulates the baseband spectrum, shifting it to be centered at $\pm\omega_0$. The upper edge of the new frequency band will be $\omega_0 + \omega_{max} = \omega_0 + \omega_s/2$.
Nyquist Rate: $2(\omega_0 + \omega_s/2) = 2\omega_0 + \omega_s$.

(c) Z-Transform of $h[n]$ and Causal Sequence Verification

Problem: Obtain the closed form expression for the Z-transform $H(z)$ of $h[n] = |n|(\frac{1}{2})^{|n|}$. State poles, multiplicity, and ROC. Which functions of z could be the Z-transform of a causal sequence? (i) $X(z) = \frac{d}{dz} \frac{(1-2z^{-1})^2}{(3-2z^{-1})^2}$ (ii) $X(z) = \frac{(z-1)^3}{(z^{-1}-1/4)^2}$.

Solution:

1. Z-Transform of $h[n]$:

Split $h[n]$ into causal and anti-causal parts:

$$h[n] = n \left(\frac{1}{2}\right)^n u[n] + (-n) \left(\frac{1}{2}\right)^{-n} u[-n-1]$$

Let $a = 0.5$. The Z-transform of $na^n u[n]$ is $\frac{az^{-1}}{(1-az^{-1})^2}$ for $|z| > |a|$. The Z-transform of $-na^{-n} u[-n-1]$ is $-\frac{az}{(1-az)^2}$ for $|z| < 1/|a|$. Thus, combining them:

$$H(z) = \frac{0.5z^{-1}}{(1-0.5z^{-1})^2} - \frac{0.5z}{(1-0.5z)^2} = \frac{0.5z}{(z-0.5)^2} - \frac{0.5z}{(1-0.5z)^2}$$

ROC: The intersection of $|z| > 0.5$ and $|z| < 2$. Thus, $0.5 < |z| < 2$.

Poles: $z = 0.5$ (multiplicity 2) and $z = 2$ (multiplicity 2).

2. Causal Sequence Verification:

For a rational function to be the Z-transform of a causal sequence, the limit as $z \rightarrow \infty$ must be finite (meaning the degree of the numerator polynomial in z cannot exceed the degree of the denominator polynomial in z).

- (i) $X(z) = \frac{d}{dz} \left[\left(\frac{1-2z^{-1}}{3-2z^{-1}} \right)^2 \right]$. The term inside the bracket goes to $(1/3)^2 = 1/9$ as $z \rightarrow \infty$. Its derivative with respect to z will thus go to 0 as $z \rightarrow \infty$. Since $X(\infty) = 0$ (which is finite), **this can be the Z-transform of a causal sequence.**
- (ii) $X(z) = \frac{(z-1)^3}{(z^{-1}-1/4)^2}$. Convert to positive powers of z : Multiply numerator and denominator by z^2 :

$$X(z) = \frac{z^2(z-1)^3}{(1-0.25z)^2}$$

The numerator is degree 5 in z , and the denominator is degree 2 in z . As $z \rightarrow \infty$, $X(z) \rightarrow \infty$. Since it is not finite, **this cannot be the Z-transform of a causal sequence** (it represents a non-causal sequence with terms extending to $n < 0$).

Question 2

(a) Cascade System Impulse Response (OR Option)

Problem: The impulse responses of two DTLTI systems are $g[n] = \{1, 8, 2, -1\}$ and $h[n] = \{1, -5, 2, -1\}$. Find the impulse response of the two systems in cascade.

Solution:

The cascade impulse response is the linear convolution $y[n] = g[n] * h[n]$. Using polynomial multiplication (or the tabular method):

$$\begin{aligned} y[0] &= (1)(1) = 1 \\ y[1] &= (1)(-5) + (8)(1) = -5 + 8 = 3 \\ y[2] &= (1)(2) + (8)(-5) + (2)(1) = 2 - 40 + 2 = -36 \\ y[3] &= (1)(-1) + (8)(2) + (2)(-5) + (-1)(1) = -1 + 16 - 10 - 1 = 4 \\ y[4] &= (8)(-1) + (2)(2) + (-1)(-5) = -8 + 4 + 5 = 1 \\ y[5] &= (2)(-1) + (-1)(2) = -2 - 2 = -4 \\ y[6] &= (-1)(-1) = 1 \end{aligned}$$

Result: $y[n] = \{1, 3, -36, 4, 1, -4, 1\}$.

(b) Inverse Z-Transform Verification

Problem: $F(z) = \log(1 - \frac{1}{2}z^{-1})$ for $|z| > \frac{1}{2}$. Verify whether or not $f[n] = -\frac{1}{n}(\frac{1}{2})^n u[n-1]$.

Solution:

We expand $F(z)$ using the standard Taylor series expansion for the natural logarithm, $\log(1 - x) = -\sum_{n=1}^{\infty} \frac{x^n}{n}$ for $|x| < 1$. Here, $x = \frac{1}{2}z^{-1}$. The convergence condition $|x| < 1 \implies |\frac{1}{2}z^{-1}| < 1 \implies |z| > 0.5$, which matches the given ROC. Substituting x into the series:

$$F(z) = -\sum_{n=1}^{\infty} \frac{1}{n} \left(\frac{1}{2}z^{-1}\right)^n = \sum_{n=1}^{\infty} \left[-\frac{1}{n} \left(\frac{1}{2}\right)^n\right] z^{-n}$$

By definition, the Z-transform is $F(z) = \sum_{n=-\infty}^{\infty} f[n]z^{-n}$. Equating the coefficients of z^{-n} , we find that for $n \geq 1$:

$$f[n] = -\frac{1}{n} \left(\frac{1}{2}\right)^n$$

And $f[n] = 0$ for $n < 1$. This can be written compactly using the unit step function as:

$$f[n] = -\frac{1}{n} \left(\frac{1}{2}\right)^n u[n-1]$$

Conclusion: The statement is **verified** and true.

(c) Stability Range of IIR System

Problem: $H(z) = \frac{1}{1+az^{-1}+0.3z^{-2}}$. Determine the range of values of a for which the causal system will be stable.

Solution:

For a causal 2nd-order IIR system with denominator $D(z) = 1 + a_1z^{-1} + a_2z^{-2}$ to be strictly stable, the poles must lie inside the unit circle. This is guaranteed by the Jury Stability Criterion (or Schur-Cohn conditions) which requires three conditions to be met simultaneously:

1. $|a_2| < 1$ 2. $1 + a_1 + a_2 > 0$ 3. $1 - a_1 + a_2 > 0$ Here, $a_1 = a$ and $a_2 = 0.3$.

- Condition 1: $|0.3| < 1$ (This is inherently satisfied).
- Condition 2: $1 + a + 0.3 > 0 \implies a > -1.3$.
- Condition 3: $1 - a + 0.3 > 0 \implies 1.3 > a \implies a < 1.3$.

Result: The system is stable for the range $-1.3 < a < 1.3$.

Question 3(b)

Inverse Z-Transform via Partial Fraction Expansion

Problem: $G(z) = \frac{z^4 - \frac{9}{4}z^3 + \frac{7}{2}z^2 - \frac{25}{4}z}{z^2 - \frac{13}{4}z + \frac{3}{4}}$. Identify all possible ROCs, find closed form expressions for $g[n]$, and comment on causality and stability.

Solution:

First, factor a z from the numerator to formulate $G(z)/z$:

$$\frac{G(z)}{z} = \frac{z^3 - 2.25z^2 + 3.5z - 6.25}{z^2 - 3.25z + 0.75}$$

The roots of the denominator $z^2 - 3.25z + 0.75 = 0$ are: $z_1 = 3$ and $z_2 = 0.25$.

Since the numerator degree (3) is greater than the denominator degree (2), perform long division:

$$\frac{z^3 - 2.25z^2 + 3.5z - 6.25}{z^2 - 3.25z + 0.75} = (z + 1) + \frac{6z - 7}{(z - 3)(z - 0.25)}$$

Perform Partial Fraction Expansion on the remainder:

$$\frac{6z - 7}{(z - 3)(z - 0.25)} = \frac{A}{z - 3} + \frac{B}{z - 0.25}$$

- $A = \frac{6(3) - 7}{3 - 0.25} = \frac{11}{2.75} = 4$
- $B = \frac{6(0.25) - 7}{0.25 - 3} = \frac{1.5 - 7}{-2.75} = \frac{-5.5}{-2.75} = 2$

Thus, $\frac{G(z)}{z} = z + 1 + \frac{4}{z - 3} + \frac{2}{z - 0.25}$. Multiplying by z :

$$G(z) = z^2 + z + \frac{4z}{z - 3} + \frac{2z}{z - 0.25}$$

The poles are at $r_1 = 3$ and $r_2 = 0.25$. This creates three possible ROCs: **ROC 1:** $|z| > 3$ (**Causal, Unstable**) Since the ROC is exterior, both fractional terms yield right-sided sequences. The unit circle is not included, making it unstable.

$$g[n] = \delta[n + 2] + \delta[n + 1] + 4(3)^n u[n] + 2(0.25)^n u[n]$$

ROC 2: $0.25 < |z| < 3$ (**Non-Causal, Stable**) The ROC includes the unit circle $|z| = 1$, making the system strictly stable. The pole at 3 yields a left-sided sequence, and the pole at 0.25 yields a right-sided sequence.

$$g[n] = \delta[n + 2] + \delta[n + 1] - 4(3)^n u[-n - 1] + 2(0.25)^n u[n]$$

ROC 3: $|z| < 0.25$ (**Anti-Causal, Unstable**) The ROC is interior, so both fractional terms yield left-sided sequences. The unit circle is not included, so it is unstable.

$$g[n] = \delta[n + 2] + \delta[n + 1] - 4(3)^n u[-n - 1] - 2(0.25)^n u[-n - 1]$$

Question 4

(a) Butterworth Filter using Impulse Invariance

Problem: Low pass Butterworth, DC gain 0 dB, $A_p = 3$ dB at 2 kHz, $A_s \geq 15$ dB at 6 kHz. $F_s = 12$ kHz. Use impulse invariant transformation to obtain the difference equation.

Solution:

1. Analog Frequencies:

For Impulse Invariant Transformation, we directly map analog frequencies $\Omega = 2\pi f$. $\Omega_p = 2\pi(2000) = 4000\pi$ rad/s.

$\Omega_s = 2\pi(6000) = 12000\pi$ rad/s.

2. Filter Order (N):

$$N \geq \frac{\log_{10} \sqrt{(10^{1.5} - 1)/(10^{0.3} - 1)}}{\log_{10}(\Omega_s/\Omega_p)} = \frac{\log_{10} \sqrt{30.62/0.995}}{\log_{10}(3)} = 1.56 \implies N = 2$$

Because $A_p = 3$ dB, we can set the cutoff frequency exactly at the passband edge: $\Omega_c = \Omega_p = 4000\pi \approx 12566.37$ rad/s.

3. **Analog Poles:** For $N = 2$, poles are $p_{1,2} = \Omega_c(-\frac{1}{\sqrt{2}} \pm j\frac{1}{\sqrt{2}}) = -8885.76 \pm j8885.76$.

4. **Digital Poles (Impulse Invariant):** $p_1^z = e^{p_1/F_s} = e^{(-8885.76 + j8885.76)/12000} = e^{-0.7405} e^{j0.7405} = 0.4769(\cos(0.7405) + j \sin(0.7405)) \approx 0.352 + j0.322$. $p_2^z = 0.352 - j0.322$.

5. **Residues and Z-Domain Transfer Function:** $H(s) = \frac{\Omega_c^2}{(s-p_1)(s-p_2)} = \frac{R_1}{s-p_1} + \frac{R_2}{s-p_2}$
 $R_1 = \frac{\Omega_c^2}{p_1 - p_2} = -j8885.76$, $R_2 = j8885.76$. Using the standard IIT: $H(z) = \frac{R_1}{1-p_1^z z^{-1}} + \frac{R_2}{1-p_2^z z^{-1}}$.
Combining them yields $H(z) = \frac{\text{num}_1 z^{-1}}{1 + \text{den}_1 z^{-1} + \text{den}_2 z^{-2}}$. Evaluating the terms: $\text{den}_1 = -(p_1^z + p_2^z) = -2(0.352) = -0.704$ $\text{den}_2 = p_1^z p_2^z = (0.352)^2 + (0.322)^2 = 0.2274$ $\text{num}_1 = -(R_1 p_2^z + R_2 p_1^z) = 5717.58$ So, the unscaled response is $H(z) = \frac{5717.58 z^{-1}}{1 - 0.704 z^{-1} + 0.2274 z^{-2}}$. To achieve a DC gain of 0 dB ($H(1) = 1$): $H(1)_{\text{unscaled}} = \frac{5717.58}{1 - 0.704 + 0.2274} \approx 10923.9$. Multiplying by scaling factor $K = 1/10923.9$:

$$H_{\text{scaled}}(z) = \frac{0.5234 z^{-1}}{1 - 0.704 z^{-1} + 0.2274 z^{-2}}$$

Difference Equation:

$$y[n] = 0.704y[n-1] - 0.2274y[n-2] + 0.5234x[n-1]$$

(b) Bilinear Warping and Characteristic Aliasing

Problem: Prove the warping formulae, point out significance, and explain why bilinear is free from characteristic aliasing.

Solution:

The Bilinear mapping is $s = \frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}}$. Setting $s = j\Omega$ and $z = e^{j\omega}$:

$$j\Omega = \frac{2}{T} \frac{1 - e^{-j\omega}}{1 + e^{-j\omega}} = \frac{2}{T} \frac{e^{-j\omega/2}(e^{j\omega/2} - e^{-j\omega/2})}{e^{-j\omega/2}(e^{j\omega/2} + e^{-j\omega/2})} = \frac{2}{T} \frac{2j \sin(\omega/2)}{2 \cos(\omega/2)} = j \frac{2}{T} \tan(\omega/2)$$

Thus, $\Omega = \frac{2}{T} \tan(\omega/2)$. **Significance:** This formula reveals a non-linear mapping (warping) between analog Ω and digital ω . To achieve a desired digital cutoff frequency ω_c , one must "pre-warp" the analog design frequency to Ω_c to account for this distortion before designing the prototype. **Absence of Characteristic Aliasing:** The tangent function uniquely maps the entire infinite imaginary axis ($-\infty < \Omega < \infty$) onto a single revolution of the unit circle ($-\pi < \omega < \pi$). Because there is a strict 1-to-1 correspondence without overlapping frequency bands, the severe aliasing inherent in the impulse invariant method is completely avoided.

Question 5 (Short Notes)

(a) Frequency spectra of uniformly sampled signals:

When an analog signal $x(t)$ with spectrum $X(f)$ is sampled uniformly at rate F_s , the discrete spectrum becomes a periodic sum of shifted copies of $X(f)$: $X_s(f) = F_s \sum X(f - kF_s)$. If the signal is not bandlimited to $F_s/2$, adjacent spectral copies will overlap, leading to destructive interference known as aliasing.

(b) Mapping of left half of s-plane on to z-plane:

In the standard mapping $z = e^{sT}$, an analog pole $s = \sigma + j\Omega$ maps to a radius $|z| = e^{\sigma T}$ and angle $\angle z = \Omega T$. The left half of the s-plane represents strictly stable poles where $\sigma < 0$. If $\sigma < 0$, then $e^{\sigma T} < 1$, which strictly forces $|z| < 1$. Thus, the entire stable left half of the s-plane maps into the interior of the unit circle on the z-plane, perfectly preserving system stability.

(c) Recursive and non-recursive systems and their frequency responses:

Recursive systems compute the current output using past outputs (feedback), typically realizing IIR filters. Their frequency response usually features highly non-linear phase but can achieve sharp magnitude roll-offs with lower orders. Non-recursive systems rely strictly on current and past inputs (FIR filters). Because they lack feedback, they are unconditionally stable and can be effortlessly designed to have perfectly linear phase frequency responses, averting any phase distortion.